① konversi:

a. $100101010,001_{(2)}$ = ... (10)

$(0\times2^{0})+(1\times2^{1})+(0\times2^{2})+(1\times2^{3})+(0\times2^{4})+(1\times2^{5})+(0\times2^{6})+(0\times2^{7})+(0\times2^{8})$

0 2 0 8 0 32 0 0 256 : 256 + 32 + 8 + 2 = 298

$(0\times2^{-1})+(0\times2^{-2})+(1\times2^{-3})$

0 0 1

0,5 0,25 0,125 → $298,125_{(10)}$

b. $2CF,8_{(16)}$

$(2\times16^{2})+(12\times16^{1})+(15\times16^{0})+(8\times16^{-1})$:

512 + 192 + 15 + 0,5 : 719,5

12 / 16 / 72 / 12 / 192

16 / 16 / 96 / 16 / 256 ×2 / 2 / 512

8 $\frac{1}{16^{2}}$ = 0,5

e. $1010101111010101_{(2)}$: ... (16)

0101 = 5

1111 : 15 : F

1010 : 10 : A

0010 : 2

2AF5 (16)

1810 / 3 | 5432 / 24 / 24 / 03 / 3 / 20

3945 / 2 | 7890 / 78 / 09 / 8 / 10

c. $456_{(8)}$: ... (16)

6 → 110

5 → 101

4 → 100

} 100101110 (2)

.0001 → 1

.0010 → 2

.1110 → 14 = E

} $12E_{(16)}$

d. $7890_{(10)}$: ... (2)

7890 : 2 = 3945 sisa 0

3945 : 2 Sisa 1

1972 : 2 sisa 0

986 : 2 sisa 0

493 : 2 sisa 1

246 : 2 sisa 0

123 : 2 sisa 1

61 : 2 sisa 1

30 : 2 sisa 0

15 : 2 sisa 1

7 : 2 sisa 1

3 : 2 sisa 1

1 : 2 Sisa 1

} $1111011010010_{(2)}$

039045 / 2 | 7890 / 0 / 78 / 78 / 09 / 0 / 90

ANEH

390 / 2 | 7890 / 78 / 09

1972 / 2 | 3945 / 19 / 18 / 14 / 14 / 051 / 1 / 10 / 14

986 / 2 | 1972 / 18 / 17 / 16 / 12

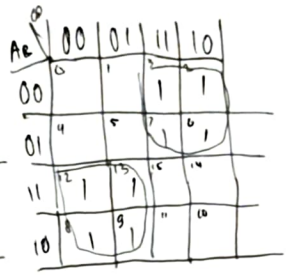
493 / 2 | 986 / 8 / 18 / 18 / 6

246 / 2 | 493 / 4 / 9 / 8 / 13

2. $Y(A,B,C,D) = \sum m\,(2,3,6,7,8,9,12,13)$

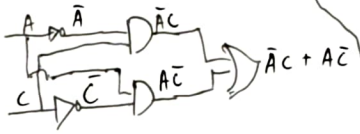
Tabel kebenaran

Minterm	A	B	C	D	Y
0	0	0	0	0	
1	0	0	0	1	
2	0	0	1	0	1
3	0	0	1	1	1
4	0	1	0	0	
5	0	1	0	1	
6	0	1	1	0	1
7	0	1	1	1	1
8	1	0	0	0	1
9	1	0	0	1	1
10	1	0	1	0	
11	1	0	1	1	
12	1	1	0	0	1
13	1	1	0	1	1
14	1	1	1	0	
15	1	1	1	1	

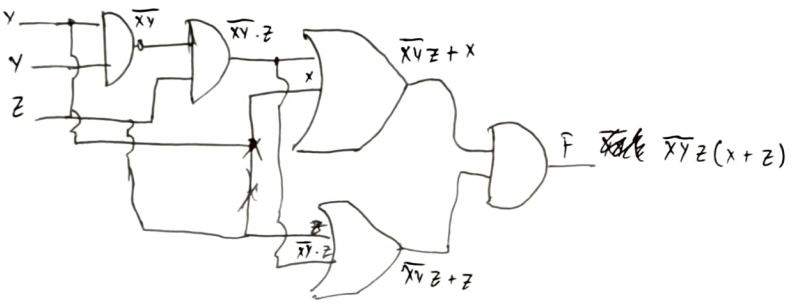


$\bar{A}C + A\bar{C}$

Skema



3.



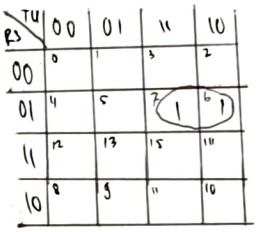
X	Y	Z	XY	$\overline{XY}$	$\overline{XY}Z$	X + Z	$\overline{XY}Z(X+Y)$
0	0	0	0	1	0	0	0
0	0	1	0	1	1	1	1
0	1	0	0	1	0	0	0
0	1	1	0	1	1	1	1
1	0	0	0	1	0	1	0
1	0	1	0	1	1	1	1
1	1	0	1	0	0	1	0
1	1	1	1	0	0	1	0

5. $F(R,S,T,U) =$

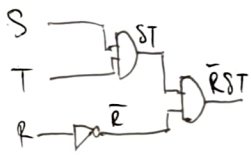
Minterm	R	S	T	U	F
0	0	0	0	0	
1	0	0	0	1	
2	0	0	1	0	
3	0	0	1	1	
4	0	1	0	0	
5	0	1	0	1	
6	0	1	1	0	1
7	0	1	1	1	1
8	1	0	0	0	
9	1	0	0	1	
10	1	0	1	0	
11	1	0	1	1	
12	1	1	0	0	
13	1	1	0	1	
14	1	1	1	0	
15	1	1	1	1	

ST aktif

R harus mati } $m(6,7)$



$\bar{R}ST = F$



1). $Y(A,B,C) = \sum m(2,3,4,6)$

a. Truth Table

Minterm	A	B	C	Y
0	0	0	0	
1	0	0	1	
2	0	1	0	1
3	0	1	1	1
4	1	0	0	1
5	1	0	1	
6	1	1	0	1
7	1	1	1	

b. Penyederhanaan SOP

$= \bar{A}B\bar{C} + \bar{A}BC + A\bar{B}\bar{C} + AB\bar{C}$

$= \bar{A}B(\bar{C}+C) + A\bar{C}(\bar{B}+B)$

$= \bar{A}B(1) + A\bar{C}(1)$

$= \bar{A}B + A\bar{C}$

